

100 Days of Azure DevOps

Day 24 Handout — restore, build, test, publish — skip one and production finds it

3 - Continuous Integration · Personal lab only · Educational content · Not a sales pitch

Architecture A — .NET CI path on a hosted agent

Step	Command / task	Output
SDK pin	UseDotNet@2 packageType sdk version 8.x	dotnet --version on ubuntu-latest
Restore	dotnet restore src/SampleApi/SampleApi.csproj	NuGet assets; can use day09-packages later
Build	dotnet build -c Release --no-restore	bin/Release/net8.0
Test	dotnet test -c Release --no-build --logger trx	trx on Test Results tab if published
Publish app	dotnet publish -c Release -o \$(Build.ArtifactStagingDirectory)/api	Runnable output, not the full src tree
Publish artifact	PublishPipelineArtifact@1 artifactName drop	Consumable by a later deploy stage
Trigger filter	trigger.paths.include: src/SampleApi/**	Do not rebuild on docs-only commits

Architecture B — decide SDK, configuration, artifact shape

Choice	Lab default	Change when
SDK	8.x via UseDotNet@2 (not whatever the image has today)	You pin 9.x globally.json to match
Configuration	Release for CI	Debug only for local F5
Artifact	Pipeline artifact drop of publish output	NuGet pack to Azure Artifacts (Day 9 feed) later
Test fail	Fail the job (no true)	You are hiding a red build
windows-latest vs ubuntu	ubuntu-latest for this API	You need Windows-only workloads (WPF, some SDKs)
Restore source	nuget.org (hosted)	Private feed: NuGetAuthenticate@1 + nuget.config

Architecture C — pitfalls that publish bin/ instead of the app

Pitfall	What you see	Fix
PublishBuildArtifacts of bin/Release	Includes obj noise, missing deps.json layout	dotnet publish -o staging then artifact that folder
true on dotnet test	Green badge, failing tests	Remove true; fix tests
No UseDotNet@2	SDK mismatch vs global.json	Pin 8.x (or your pin) on the job
Path trigger too wide	README typo burns hosted minutes	paths.include src/SampleApi/**
Restore without --no-restore later	Double restore, slower CI	restore once; build --no-restore
Checking in bin/	PRs unreviewable	Day 13 gitignore; git rm --cached

One-liner to remember

.NET CI is the same ritual every time: restore, build, test, publish — skip one and production finds it for you.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal azure-100-labs. Create a tiny Web API under /src/SampleApi. Microsoft-hosted ubuntu-latest. No work NuGet feeds.

- 1 Local SDK 8: dotnet new webapi -n SampleApi -o src/SampleApi && dotnet new xunit -n SampleApi.Tests -o src/SampleApi.Tests && dotnet add src/SampleApi.Tests/SampleApi.Tests.csproj reference src/SampleApi/SampleApi.csproj. One Assert.True is enough.
- 2 Commit on feature/day24-dotnet-ci. Do not commit bin/ obj/.
- 3 Add pipelines/dotnet-ci.yml (code below). Pipelines → New pipeline → Existing YAML → this file. Or extend azure-pipelines.yml with a path filter.
- 4 Push a PR. Confirm UseDotNet@2 logs SDK 8.x, restore, build, test, publish artifact drop.
- 5 Open the run → Artifacts → drop. Confirm published output, not the whole repo.
- 6 If tests were skipped, add at least one test project before calling CI done. Do not leave || true.
- 7 Squash merge when green. Hosted pool only.

pipelines/dotnet-ci.yml — restore, build, test, publish drop

```
trigger:
  branches:
    include:
      - main
  paths:
    include:
      - src/SampleApi/**
      - src/SampleApi.Tests/**
      - pipelines/dotnet-ci.yml
pr:
  branches:
    include:
      - main
  paths:
    include:
      - src/SampleApi/**
      - src/SampleApi.Tests/**
      - pipelines/dotnet-ci.yml

pool:
  vmImage: ubuntu-latest

steps:
- task: UseDotNet@2
  inputs:
    packageType: sdk
    version: 8.x

- script: |
  dotnet restore src/SampleApi.Tests/SampleApi.Tests.csproj
  dotnet build src/SampleApi.Tests/SampleApi.Tests.csproj -c Release --no-restore
  dotnet test src/SampleApi.Tests/SampleApi.Tests.csproj -c Release --no-build --logger trx
  dotnet publish src/SampleApi/SampleApi.csproj -c Release --no-build -o $(Build.ArtifactStagingDirectory)/api
i
  displayName: restore, build, test, publish

- task: PublishPipelineArtifact@1
  inputs:
    targetPath: $(Build.ArtifactStagingDirectory)/api
    artifact: drop
```

Done checklist

- src/SampleApi exists and builds on ubuntu-latest
- Pipeline pins SDK 8.x with UseDotNet@2
- Artifact drop is publish output, not raw bin/
- No || true hiding test failures — red tests fail the job

Tomorrow — Day 25

CI for Node.js

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